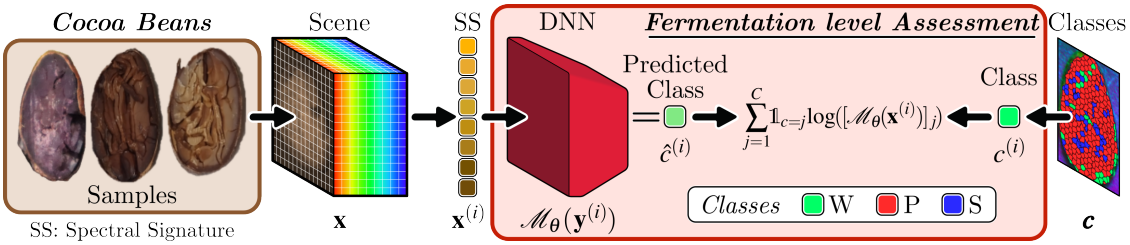
Spectral Assessment for Cocoa Quality

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Abstract

This work presents a method combining near-infrared (NIR) spectroscopy with machine learning to assess the quality of Colombian cocoa beans, offering a fast and non-invasive alternative to traditional, subjective, and destructive methods. Cocoa quality is primarily determined by the fermentation level. Therefore, this work adapt different models to enhance the spectral data analysis, promoting sustainable and reliable practices in the Colombian cocoa industry.

Near-infrared for cocoa bean quality estimation

The NTC-1252-2021 standard in Colombia assesses cocoa quality by evaluating the fermentation of 100 beans, which are cut and visually classified by an expert as well-fermented (W), partially fermented (P), or slightly-fermented (S). Traditional methods destroys the beans to make a chemical based assessment. In contrast, NIR spectroscopy captures spectral data that can be used to estimate fermentation levels through machine and deep learning methods in a non-destructive way to estimate fermentation levels through machine and deep learning methods [1].

Learning-Based

Given a dataset $\{\mathbf{x}^{(i)}, c^{(i)}\}_{i=1}^{S}$ with S spectral samples $\mathbf{x}^{(i)}$ and corresponding labels $c^{(i)} \in \{1,2,3\}$ representing the fermentation levels (W, P, S), we aim to optimize a deep neural network (DNN) $\mathscr{M}_\theta(\cdot) \in \mathbb{R}$ with parameters θ. The overview of the proposed method is shown in the figure, whose optimization is

$$\theta^* \in \arg\min_{\theta} -\sum_{i=1}^{S}\sum_{j=1}^{C} \mathbb{1}_{c=j} \log([\mathscr{M}_\theta(\mathbf{x}^{(i)})]_j), \tag{1}$$

where $\mathbb{1}_v$ is an identity function that is equal to 1 if v is true and 0 otherwise, and $[\mathscr{M}_\theta(\cdot)]_j$ is the jth element of the network output, corresponding to the predicted probability for class j.

Results

For evaluation, we used the dataset provided in [1] and five different methods, where Spectralformer [3] outperformed all methods, achieving an OA metrics of 97.53%, an AA of 98.39%, and a Kappa score of 93.28%. It performed 5.9% better in OA than SVM, which highlights Spectralformer's superior ability to capture long-range dependencies in spectral data.

Method	OA	AA	Kappa
SVM	92.09 ± 0.51	94.93 ± 0.55	92.58 ± 0.78
MLP	92.00 ± 1.02	92.56 ± 0.77	87.99 ± 1.84
CNN	96.47 ± 0.15	97.48 ± 0.18	92.68 ± 0.24
SpectralNET [2]	94.05 ± 2.13	94.95 ± 1.60	90.95 ± 3.24
Spectralformer [3]	**97.53 ± 0.77**	**98.39 ± 0.81**	**93.28 ± 1.18**

Acknowledgements: *This work was supported by the VIE-UIS under project "Apoyo a semilleros de investigación - Diseño de codificación para el muestreo compresivo de señales multidimensionales utilizando técnicas basadas en aprendizaje profundo - 4159" and project 3924.*

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